Herbert William George MacLeod, B.Sc., M.D., M.R.C.P., D.P.H., 20 Craven Hill, Lancaster Gate, London, W. 2 (proposed by W. A. Parr); and

Bruno Philip Henry de Roeper, Squadron Leader R.A.F., No. 1 Flying Training School, Netheravon, Wilts (proposed by W. J. S. Lockyer).

One hundred and ninety-three presents were announced as having been received since the last meeting, including, amongst others:—

Tycho Brahe, Opera Omnia, Tomus VII. (presented by Dr. Dreyer).

British (Terra Nova) Antarctic Expedition, 1910–1913:—C. Chree, Terrestrial Magnetism; H. G. Lyons, Miscellaneous Data; G. C. Simpson, Meteorology (presented by the Captain Scott Antarctic Fund).

Carte photographique du ciel: Zone $+14°$, Bordeaux Observatory (5 charts); Zones $-3°$ and $-5°$, San Fernando Observatory (31 charts): Zones $+5°$ and $+7°$, Toulouse Observatory (12 charts).

H. Dingle, Modern Astrophysics.

C. Lane Poor, Gravitation versus Relativity.

Washington, Hydrographic Office, Publication No. 203: The Sumner Line of Position (presented by Mr. H. B. Goodwin).

Mount Wilson Observatory, Seven Transparencies of Nebulæ and Portions of the Moon, taken with the 60-inch and 100-inch Reflectors.

Sproul Observatory, Swarthmore, Two Transparencies of the Total Solar Eclipse of 1923 September 10, taken at Yerbanis, Mexico.

G. Percy Wilkins, Twenty-four Charts of the Moon, with Index Map.

J. H. Worthington, Ten Transparencies of the Eclipse of 1923 September 10.

Cosmogonic Problems associated with a Secular Decrease of Mass.

By J. H. Jeans, D.Sc., Sec. R.S.

1. Some years ago I put forward the suggestion that the energy of stellar radiation is produced by a secular decrease in the star's mass, the mechanism possibly being that positive and negative electric charges fall together and annihilate one another, their energy being transformed into radiation. This conjecture has gained enormously in probability since Eddington has shown * that, as a matter both of theory and of observation, the radiation from a star is approximately a function only of the star's mass. It is now clear, both from theory and observation, that as a star's development proceeds, its mass must decrease. The star cannot, in any way known to us, store up more than a quite insignificant fraction of the energy liberated by this destruction of mass, so that it is

* *M.N.*, **84** (1924), 308.

natural to suppose that this energy escapes as radiation, the more so as this supposition solves the otherwise insoluble problem of the source of the energy of stellar radiation. Quite independently of the nature of the mechanism by which mass is transformed into radiation, the theory of relativity fixes the rate of exchange between these two forms of energy. The destruction of mass m sets free energy mC^2 where C is the velocity of light, this energy now being free to escape as radiation. The converse is also true; if energy E escapes as radiation, the mass of the radiator must decrease by E/C^2. For instance, from the known value of the solar constant, we can calculate that the sun must be losing mass at the rate of 4,200,000 tons a second. The mass of a giant star of absolute magnitude -5 must be decreasing at 10,000 times this rate.

Discussion of the problems of cosmogony has hitherto proceeded on the supposition that each star retains a constant mass throughout the whole course of its development. The conception of a secular decrease of mass brings new features into most of these problems; it is worth examining whether these carry with them the answers to any of the unsolved problems of cosmogony.

For instance, H. N. Russell has shown in a well-known paper * that, so long as outside disturbances are absent, the orbits of binary stars can only increase in size to a very limited extent. In obtaining this result, it was implicitly assumed that the masses of the components remain constant. If these masses are continually decreasing, the orbit can increase to a far greater extent than was calculated by Russell. This is best seen by passing to the limiting case in which both masses are suddenly reduced to zero; the orbits now reduce to straight lines which may be regarded as parabolas, or as ellipses of unlimited size. Among other problems, we shall have to examine whether this hitherto unconsidered possibility is capable of solving the mystery of the observed distribution of the orbits of binary stars.†

2. By an artifice as old as Newton, the problem of the gravitational motion of two bodies can be reduced to that of the motion of a particle about a fixed centre of force. It is readily seen that the same method can be used when the masses are in process of shrinkage; to determine the motion of either, we study the motion of a particle under a central force M/r^2, where $M = m^3/(m+m')^2$, m, m' being the masses of the two components whether constant or changing.

The motion of the particle will be governed by the usual equations

$$r^2\dot{\theta} = n \quad . \quad . \quad . \quad . \quad . \quad (1)$$

$$\ddot{r} - r\dot{\theta}^2 = -M/r^2 \quad . \quad . \quad . \quad . \quad (2)$$

in which n is a constant but M varies with the time. From these equations we readily deduce

$$\frac{d}{dt}\left[\tfrac{1}{2}(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{M}{r}\right] = -\frac{1}{r}\frac{dM}{dt} \quad . \quad . \quad . \quad (3)$$

* *Astrophys. Journ.*, **31** (1910), 185.

† This possibility was suggested to me by Professor Russell.

showing that the energy of the particle in its orbit does not remain constant. Since dM/dt for an actual star will always be negative, the energy of the particle in its orbit continually increases. The rate of increase is represented by the right-hand member of equation (3) multiplied by the mass of the particle; this is equal to the rate at which work is done in removing the radiation emitted by the star out of the gravitational field of the particle.

3. From equations (1) and (2) we obtain in the usual way

$$\frac{d^2r}{dt^2}-\frac{n^2}{r^3}=-\frac{M}{r^2} \quad . \quad . \quad . \quad . \quad . \quad (4)$$

an equation which does not admit of direct integration unless M is completely specified as a function of t.

Putting $Mr=\lambda$, the equation transforms into

$$\frac{1}{M^3}\frac{d^2}{dt^2}\left(\frac{\lambda}{M}\right)=\frac{n^2}{\lambda^3}-\frac{1}{\lambda^2} \quad . \quad . \quad . \quad . \quad . \quad (5)$$

If we now put

$$q=\int M^2dt,$$

so that q is a function of t only, and

$$\frac{d}{dt}=M^2\frac{d}{dq},$$

we find that equation (5) becomes

$$\frac{d^2\lambda}{dq^2}-\frac{\lambda}{M}\frac{d^2M}{dq^2}=\frac{n^2}{\lambda^3}-\frac{1}{\lambda^2} \quad . \quad . \quad . \quad . \quad . \quad (6)$$

For this to be directly integrable we must have

$$\frac{d^2M}{dq^2}=\alpha^2M \quad . \quad . \quad . \quad . \quad . \quad . \quad (7)$$

where α^2 is a constant. Integrating equation (7) with respect to M and restoring t as variable, we find that equation (7) requires that

$$\frac{dM}{dt}=-M^2(\alpha^2M^2+\beta)^{\frac{1}{2}} \quad . \quad . \quad . \quad . \quad . \quad (8)$$

where β is a constant of integration. The total rate of radiation L is, of course, connected with the rate of diminution of mass by the relation

$$L=-C^2\frac{dM}{dt} \quad . \quad . \quad . \quad . \quad . \quad . \quad (9)$$

so that equation (6) becomes integrable if the emission L is proportional to $M^2(\alpha^2M^2+\beta)^{\frac{1}{2}}$.

4. According to Eddington's theory, the emission L from stars of very great mass is proportional to $M^{1\cdot4}$, while that for stars of very small mass is proportional to $M^{4\cdot4}$. For stars of intermediate mass, the index of M is intermediate between the two extreme values of 1·4 and 4·4,

and Eddington's table * shows that there is a very considerable range of mass, corresponding roughly to the range between absolute magnitudes 5 and − 1, for which the emission varies approximately as M^3. This can be represented by taking $\beta = 0$ in equation (8), so that

$$\frac{dM}{dt} = -\alpha M^3 \quad . \quad . \quad . \quad . \quad . \quad . \qquad (10)$$

Equation (4) now becomes

$$\frac{d^2\lambda}{dq^2} = \alpha^2\lambda + \frac{n^2}{\lambda^3} - \frac{1}{\lambda^2} \quad . \quad . \quad . \quad . \quad . \qquad (11)$$

whence, on integrating with respect to λ,

$$\frac{d\lambda}{dq} = \left(\alpha^2\lambda^2 - \frac{n^2}{\lambda^2} + \frac{2}{\lambda} + A\right) \quad . \quad . \quad . \quad . \quad . \qquad (12)$$

where A is a constant of integration.

The equation

$$\alpha^2\lambda^2 + A + \frac{2}{\lambda} - \frac{n^2}{\lambda^2} = 0 \quad . \quad . \quad . \quad . \quad . \qquad (13)$$

can have either one or three real positive roots in λ. The case of one real root corresponds to a quasi-hyperbolic orbit which is of no special cosmogonic interest. When there are three real positive roots, (if we reject the solution with λ larger than the largest root) the motion in λ determined by equation (12) will be an oscillatory motion between two adjacent roots, say $\lambda = \lambda_1$ and $\lambda = \lambda_2$. Since $\lambda = Mr$, it is clear that when the mass of the star is M the values of r will be oscillating between the values λ_1/M and λ_2/M. If M is not varying too rapidly, the orbit may be regarded as an ellipse, for which, if a, e denote its semi-major axis and eccentricity,

$$a(1-e) = \lambda_1/M$$
$$a(1+e) = \lambda_2/M.$$

From these equations

$$e = \frac{\lambda_2 - \lambda_1}{\lambda_2 + \lambda_1},$$

so that e depends only on the constants in equation (13) and so is independent of the time. We further have

$$a = \frac{\lambda_1 + \lambda_2}{2M} \quad . \quad . \quad . \quad . \quad . \quad . \qquad (14)$$

showing that, as M continually decreases, the dimensions of the orbit increase without limit, the product Ma remaining constant.

Thus the orbit of a particle is, to a first approximation, an ellipse which continually increases in size while keeping its eccentricity constant. In particular, a binary star which starts with a circular orbit must retain a circular orbit until it is disturbed by external influences. Clearly there is no explanation here of the observed high eccentricities

* *M.N.*, **84**, 310.

of binary systems, while the theoretically calculated rate of increase of the dimensions of the orbit is too slow to offer an explanation of the observed wide separations of the components of later-type binaries.

5. If the radiation of the star is not strictly proportional to M^3, or to the more general form $M^2(\alpha^2 M^2 + \beta)^{\frac{1}{2}}$, these conclusions need modification. In equation (11), α^2 may no longer be treated as a constant, so that equation (12) will not be a true integral. The value of λ will still oscillate between two values λ_1, λ_2, but these will show a secular variation with the time, so that there will be some variation in the eccentricity of the orbit.

The general problem is best attacked by returning to equation (3). If a is the semi-major axis of the elliptic orbit at any instant, the energy of the particle in its orbit is $-M/2a$, so that equation (3) becomes

$$\frac{d}{dt}\left(\frac{M}{2a}\right) = \frac{1}{r}\frac{dM}{dt}.$$

On averaging over a complete revolution of the particle, this becomes

$$\frac{d}{dt}\left(\frac{M}{2a}\right) = \frac{1}{a}\frac{dM}{dt} \quad . \quad . \quad . \quad . \quad . \quad (15)$$

it being easily shown that the average value of $1/r$ in an elliptic orbit is $1/a$. Equation (15) is now seen to have the integral

$$Ma = \text{constant} \quad . \quad . \quad . \quad . \quad . \quad (16)$$

The student of quantum-dynamics will notice that $(Ma)^{\frac{1}{2}}$, being proportional to the mean kinetic energy of the orbit divided by its frequency, is an "adiabatic invariant" and so necessarily remains constant throughout changes in M.

Thus, when the radiation is not restricted to vary as the cube of the mass, the major axis $2a$ still varies as $1/M$, while the eccentricity varies by an amount which depends on the extent to which the proportionality between M^3 and the emission fails.

6. Both from the general solution and the particular solution, it is clear that the conception of a secular decrease of mass brings no direct solution to the problem of the orbits of binary stars. But the time-scale to which this conception leads is such as almost of itself to provide a solution. Inserting the values appropriate to the sun $\left(M = 2 \times 10^{33},\ -\frac{dM}{dt} = 4{\cdot}2 \times 10^{12}\right)$, we find from the equation

$$\frac{dM}{dt} = -\alpha M^3 \quad . \quad . \quad . \quad . \quad . \quad (17)$$

that

$$\alpha = 5{\cdot}2 \times 10^{-88}.$$

Solving equation (17), it appears that the time required for a star to pass from mass M′ to mass M is given by

$$t = \frac{1}{2\alpha}\left(\frac{1}{M^2} - \frac{1}{M'^2}\right) \quad . \quad . \quad . \quad . \quad . \quad (18)$$

whence we find, if our sun ever had four times its present mass (corresponding roughly to a B-type star), the time since this period must be about $7 \cdot 1 \times 10^{12}$ years. The time since our sun had an enormously greater mass would be approximately the same, for the very massive part of the star's existence lasts only a short time. If we put $M' = \infty$ in equation (18), we obtain $t = 7 \cdot 6 \times 10^{12}$ years, but it must be remembered that our assumed law of radiation ($\propto M^3$) is not accurate for stars in the very massive state. In any case, since fission into binaries appears generally to occur somewhere about type B, or possibly earlier, we conclude that a binary which has been formed by fission and is now in the stage of development of our sun has probably been binary for something of the order of 7×10^{12} years. I have estimated * that, with the universe as it now is, a star may expect to encounter other stars at distances less than the radius of Neptune's orbit, at the rate of one encounter every 4×10^{12} years. Thus G-type dwarf binaries will on the average have experienced about two such encounters, and these would almost, although perhaps not quite, suffice to knock about the orbits of binary stars to the extent indicated by their observed eccentricities. For binaries which represent the relics of two separate condensations in the parent nebula, the argument is the same; the only difference is that we may perhaps estimate the life of the binary as somewhat longer.

It will be convenient to notice in passing some other consequences of the extended time-scale to which the conception of decreasing mass has led. With the universe as it now is, a star would expect to be deflected 7 degrees or more by encounters with another star about once in 4×10^{12} years,† so that a G-type dwarf would probably have experienced about two such encounters, as well as a number of less violent ones, in the course of its life. This goes some way at least towards explaining the tendency towards equipartition of energy which emerges from a study of the correlation between the masses and velocities of the star. Each time a light star encounters a massive one moving with a nearly equal velocity at a distance near enough to produce an appreciable deflection of orbit, the expectation is that the velocity of the massive star will be decreased, while that of the lighter star will increase. After an infinite number of such encounters there would be established the equipartition law, according to which the velocities of all the stars of any single type are dispersed about a mean which is inversely proportional to the square root of the mass. Halm has shown that the correlation between the masses and velocities of stars of different spectral types approximately conforms to this law.

Finally, it can be shown ‡ that a sun with a radius equal to that of Neptune's orbit would experience tidal pulls from passing stars of sufficient intensity to tear off planets, at the rate of about one in 3×10^{10} years. For a sun of radius equal to the earth's orbit the period is of the order of 10^{13} years. Thus we see that, even with the extended time-scale, the birth of satellites (on the tidal theory) is hardly a

* *Cosmogony and Stellar Dynamics*, p. 224.
† *Loc. cit.*, p. 226.
‡ *Loc. cit.*, p. 279.

normal event in a star's life, although it is now seen to be much less abnormal than has hitherto appeared to be the case.

7. Returning to the main problem, let γ denote the angle between the actual orbit of a particle at any point and a circular orbit through the same point, so that

$$\tan\gamma = \frac{\dot{r}}{r\dot{\theta}}.$$

Putting $r = \lambda/M$ as before, this becomes

$$\tan\gamma = \frac{1}{\dot{\theta}}\left[\frac{1}{\lambda}\frac{d\lambda}{dt} - \frac{1}{M}\frac{dM}{dt}\right] \quad . \quad . \quad . \quad (19)$$

In the particular case in which the emission is proportional to M^3, there are orbits (corresponding to circular orbits in the case of constancy of mass) for which λ retains a constant value, so that $d\lambda/dt = 0$. If we further put $n = r^2\dot{\theta} = \lambda^2\dot{\theta}/M^2$, equation (19) reduces to

$$\tan\gamma = -\frac{\lambda^2}{nM^3}\frac{dM}{dt} = \frac{a\lambda^2}{n} \quad . \quad . \quad . \quad . \quad (20)$$

by equation (17). The value of $\tan\gamma$ is accordingly constant throughout the description of the orbit, and the orbit is the equiangular spiral

$$r = Ae^{\theta\tan\gamma} \quad . \quad . \quad . \quad . \quad . \quad (21)$$

In the more general case in which the emission is not proportional to M^3, equation (19) shows that the value of γ arises from two contributions, one depending on the fluctuations in λ, and the other on the secular changes in M. If we disregard the former, equation (19) shows that the value of γ will only be appreciable when $\frac{1}{M}\frac{dM}{dt}$ is appreciable in comparison with $\dot{\theta}$; that is to say, when the time-scale for the decrease in M is not enormously large in comparison with the period of an orbit. As we pass to larger and larger masses the time-scale, as measured by the reciprocal of $\frac{1}{M}\frac{dM}{dt}$, continually decreases, while the period of an orbit may either increase or decrease. To find instances in which γ is appreciable we must look for slow orbits about enormously massive centres.

8. It is to the spiral nebulæ that we naturally look for instances of such motions. The main body of the nucleus of a spiral is transparent, so that Eddington's theoretical calculation of emission in terms of mass will not apply, but the bright central (stellar) part of the nucleus must be thoroughly opaque, and if this is gaseous the theoretical formula ought to apply. If the mass of the opaque part of the nucleus is M times that of the sun, the emission ought to vary as $M^{\frac{1}{2}}(1-\beta)^{\frac{3}{2}}$, where $(1-\beta)$ is the ratio of radiation pressure to the total pressure. This is equal to 0·05 for the sun, so that $(1-\beta)^{-\frac{3}{2}} = 89{\cdot}4$; for the massive central part of the nebular nucleus $(1-\beta)$ cannot differ appreciably

from unity. Thus we may take the radiation from the nucleus to be $89{\cdot}4 \times M^{\frac{7}{5}}$ times that of the sun.

The sun radiates $1{\cdot}9$ ergs per second for each gramme of its mass, so that the nuclear star must radiate $170\ M^{\frac{2}{5}}$ ergs per gramme per second. The energy set free by the destruction of a gramme being 9×10^{20} ergs, it follows that if the nuclear star were to continue to radiate at its present rate, the time for which it would last would be

$$\frac{M}{dM/dt} = \frac{9 \times 10^{20}}{170\ M^{\frac{2}{5}}} \text{ seconds} . \qquad (22)$$

Opik * estimates the mass of the central part of the Andromeda nebula to be $1{\cdot}8 \times 10^9$ times the mass of the sun. It is difficult to guess how much of this mass ought to be assigned to the central star; if we overestimate the mass of the central star by assigning to it the whole of this mass, the time-scale provided by equation (22) becomes

$$\frac{M}{dM/dt} = 3 \times 10^7 \text{ years} . \qquad (23)$$

Using Opik's estimate of 450,000 parsecs for the distance of the Andromeda nebula, Pease's spectroscopic velocity-determinations indicate a period of rotation at a distance of 150″ from the centre of

$$2{\cdot}3 \times 10^7 \text{ years.}$$

This period is of the same order of magnitude as that given by equation (23), so that $\tan \gamma$, as given by equation 19, will be appreciable, and the orbits under gravitation (subject to the accuracy of the estimates we have used) must be spirals of quite appreciable angle.

Unfortunately, the results we have obtained do not accord numerically with the observed brightness of the central star. The radiation from this nucleus has been taken to be $89{\cdot}4 \times M^{\frac{7}{5}}$ times that of the sun; with the admittedly excessive value which we have used for M (namely, $M = 1{\cdot}8 \times 10^9$), this factor is found to be equal to about 10^{15}. Now, a star of luminosity 10^{15} times that of the sun, at our assumed distance of 450,000 parsecs would appear of magnitude -9, whereas the central part of the nucleus of the Andromeda nebula is only of magnitude about $+6$; even if we assume the light of the outer parts to arise entirely from reflection of light emitted by the central star, the magnitude of the latter, now represented by the magnitude of the integrated light of the whole nebula, is only about $+5$.

A central star of magnitude $+5$ at a distance of 450,000 parsecs would be of absolute magnitude -9, with a luminosity only 2×10^9 times that of the sun. This seems to suggest that for the largest masses of all the radiation may be proportional simply to M instead of to $M^{\frac{7}{5}}$, as we have imagined; this could be reconciled with Eddington's theory if we could suppose that the coefficient of absorption k approximated in very great masses to a constant limiting value k_0. With this smaller value for the emission of radiation, the time-scale for

* *Astrophys. Journ.*, 55 (1922), 406.

diminution of mass approximates to that for an ordinary star and the orbits differ only imperceptibly from circles. Although the conception of a secular decrease of mass fails to explain the spiral arms of nebulæ in terms of our present knowledge of the nebulæ, the attempt has been put on record as suggesting the possibility of equiangular and other spirals being described under ordinary gravitational forces.

9. The mass of the galactic system, of which our sun is a member, must be decreasing with a time-scale which is comparable with that of a typical star. If, as Poincaré suggested, the Milky Way may be regarded as an encircling belt of stars held in equilibrium by slow rotation, then the radius of this ring must be continually expanding, being connected with the mass of the galaxy approximately by the relation $Ma = \text{constant}$. If, as is more natural, we regard the galaxy as a system of independently moving stars, the same relation $Ma =$ constant must apply to the radius of the orbit of a single star encircling the whole galaxy. Whichever way we regard the matter, it is found that the galaxy must be slowly expanding, approximately in accordance with the relation $Ma = \text{constant}$. When our sun was a B-type star, and so of mass equal to four times its present mass, the mass of the whole galaxy must have been at least four times what it now is. At this time the radius of the galaxy would be less than a quarter of its present radius, and the stars packed together at least sixty-four times as closely as they now are. Thus the chances of close encounters between stars would be sixty-four times that estimated in § 6.

More precisely, let us assume the law of shrinkage of mass for each individual star to be that already assumed in equation (17), namely:

$$\frac{dM}{dt} = -\alpha M^3.$$

If a star is of mass M_0 at time 0, its mass after time t will be

$$M = (2\alpha t + 1/M_0^2)^{-\frac{1}{2}},$$

whence it follows that the radius of the galaxy after time t will be given by

$$a = a_0(1 + 2\alpha t M_0^2)^{\frac{1}{2}}$$

and the density of stars ν by

$$\nu = \nu_0(1 + 2\alpha t M_0^2)^{-\frac{3}{2}}.$$

The total effect of the encounters of a star with its neighbours in any time from 0 to t will be proportional to

$$\int \nu dt = \frac{\nu_0}{\alpha M_0^2}\left[1 - \frac{1}{(1 + 2\alpha t M_0^2)^{\frac{1}{2}}}\right] = \frac{\nu_0}{\alpha M_0^2}\left(1 - \frac{M}{M_0}\right).$$

On further simplifying, this is found to be equal to

$$\nu t \times \frac{2M_0^2}{M(M + M_0)},$$

so that, so far as regards the effects of encounters between stars, the effect of early closer packing is to increase the time-scale by a factor

$$\frac{2M_0^2}{M(M+M_0)}.$$

If we take $M_0/M=4$, this factor becomes equal to 6·4; with $M_0/M=6$, it is equal to 10·3. If we multiply all the expectations mentioned in § 6 by the factor of 6·4 or 10·3, we find that the conception of a secular decrease of mass just about accounts for the observed orbits of binary stars and for the observed correlation between the masses of the stars and their velocities, while leaving the tidal genesis of planets a somewhat, although not excessively, rare event.

The Moon's Mean Longitude, Longitudes of Perigee and Node, Semi-Diameter and Parallactic Inequality derived from Occultations of Stars observed at the Royal Observatory, Cape of Good Hope, 1880–1922. By H. Spencer Jones, M.A., B.Sc., H.M. Astronomer.

The details of the occultations of stars by the Moon observed at the Royal Observatory, Cape of Good Hope, together with the resulting equations of condition, have been published for the period 1835 to 1880 in the *Cape Annals*, **1**, part 4; for the period 1881 to 1895 in the *Cape Annals*, **2**, part 3; and for the period 1896 to 1906 in the *Cape Annals*, **2**, part 6. The details of the observations for the period 1907 to 1922 are in manuscript ready for publication. The whole series of observations have been reduced on a uniform plan, the places of the Moon being derived from Hansen's Tables, without Newcomb's corrections. The tabular places for the first series were supplied by Newcomb.

The positions of the Moon given in the *Nautical Almanac* from 1923 onwards are based upon Brown's Tables, which will henceforth be used in the reduction of the Cape Observations of Occultations. It therefore appeared desirable to discuss the observations up to the end of 1922 with a view to deriving the Moon's longitude, semi-diameter, eccentricity of orbit, and coefficient of the principal term in the parallactic inequality. The numbers of observations in many of the years before 1880 were very few; the discussion has therefore been restricted to the period 1880 to 1922.

The equations of condition have been published in the form

$$\frac{\sin\sigma}{\sin 1''}-S=(\alpha)\Delta\alpha+(\delta)\Delta\delta+(\alpha')\Delta\alpha'+(\delta')\Delta\delta'+(T)\Delta T$$
$$+(l)\Delta l+(\phi')\Delta\phi'+(p)p+(s)s,$$